

STATISTICAL ANALYSIS OF CROP RECOMMENDATION DATASET USING R

A Data Analytics Portfolio Project

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Tools Used:

R · ggplot2 · tidyverse · corrplot · GGally · psych · skimr · janitor

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1. Executive Summary

This report presents a comprehensive statistical analysis of the Crop Recommendation dataset, conducted in R using the tidyverse ecosystem alongside specialized packages including corrrplot, GGally, psych, skimr, and janitor. The objective of the project is to explore the physicochemical and climatic conditions associated with 22 distinct crop types, characterize the statistical properties of each variable, and quantify the relationships between soil nutrients, temperature, humidity, soil pH, and rainfall.

The dataset comprises 2,200 observations across 8 variables — seven numeric predictors (nitrogen, phosphorus, potassium, temperature, humidity, soil pH, and rainfall) and one categorical target variable (crop label) spanning 22 perfectly balanced crop classes of 100 records each. The dataset was found to be exceptionally clean, containing zero missing values and zero duplicate records, which allowed the analysis to proceed directly from import into exploratory and inferential work without a remediation phase.

The analytical workflow followed a structured pipeline: data import and inspection, column-name standardization, missing-value and duplicate audits, data-type conversion of the crop label to a factor, descriptive statistics (including mean, median, standard deviation, quartiles, and coefficient of variation), univariate exploratory data analysis via histograms and boxplots, and bivariate/multivariate correlation analysis using Pearson correlation coefficients, a correlation heatmap, scatterplots with fitted regression lines, and formal significance testing of each pairwise correlation.

Key findings show that most environmental variables are only weakly correlated with one another, indicating that each contributes largely independent information to crop suitability. The single notable exception is the strong positive relationship between phosphorus (P) and potassium (K) levels ($r = 0.74$), which is statistically significant and agronomically consistent with common fertilizer and soil-management practices. Rainfall, potassium, and nitrogen each display right-skewed distributions with pronounced high-value outliers, while temperature and soil pH are more symmetric and approximately bell-shaped. These characteristics, together with the balanced class structure of the target variable, make the dataset well suited for downstream classification and crop-recommendation modeling.

2. Introduction

2.1 Importance of Statistical Analysis in Agriculture

Modern agriculture increasingly relies on data-driven decision-making to optimize yield, reduce input costs, and promote sustainable land use. Statistical analysis provides the foundation for understanding how soil nutrients, climatic variables, and management practices interact to determine crop suitability. By systematically summarizing and visualizing agricultural data, analysts can identify which environmental factors matter most, detect anomalies or data-quality issues, and generate evidence-based recommendations that support farmers, agronomists, and policymakers.

2.2 Why Crop Recommendation Datasets Are Useful

Crop recommendation datasets link measurable soil and climate conditions — nitrogen, phosphorus, potassium concentrations, temperature, humidity, pH, and rainfall — to the crops that are historically well-suited to those conditions. Such datasets serve as the backbone for precision-agriculture systems, enabling recommendation engines that suggest the most appropriate crop for a given plot based on its measured soil and weather profile. Beyond predictive modeling, they are equally valuable for exploratory statistical work that reveals the underlying structure of agronomic data, including typical nutrient ranges, distributional shape, and inter-variable relationships.

2.3 Objectives of the Project

- To import, clean, and validate the Crop Recommendation dataset for statistical analysis.
- To compute and interpret descriptive statistics for all numeric variables.
- To conduct exploratory data analysis (EDA) using histograms, boxplots, and bar charts to understand variable distributions and the balance of crop classes.
- To quantify linear relationships among numeric variables using Pearson correlation analysis, a correlation heatmap, and significance testing.
- To translate statistical findings into practical, agriculturally meaningful insights.
- To document the workflow and results in a professional, portfolio-ready format.

3. Dataset Description

The dataset used in this analysis is the publicly available Crop Recommendation Dataset, commonly hosted on Kaggle, which is widely used for agricultural machine-learning and statistical demonstration projects.

Dataset Source: Kaggle — Crop Recommendation Dataset

Number of Observations: 2,200 rows

Number of Variables: 8 columns (7 numeric, 1 categorical)

Missing Values / Duplicates: None detected (0 missing cells; 0 duplicate rows)

3.1 Variable Descriptions

Variable	Data Type	Description
n	Integer	Ratio nitrogen content in soil (kg/ha)
p	Integer	Ratio phosphorus content in soil (kg/ha)
k	Integer	Ratio potassium content in soil (kg/ha)
temperature	Numeric	Ambient temperature in degrees Celsius (°C)
humidity	Numeric	Relative humidity (%)
ph	Numeric	Soil pH value (acidity/alkalinity)
rainfall	Numeric	Rainfall in millimetres (mm)
label	Factor (22 levels)	Recommended crop type, e.g. rice, maize, coffee, mango

Each of the 22 crop classes — apple, banana, blackgram, chickpea, coconut, coffee, cotton, grapes, jute, kidneybeans, lentil, maize, mango, mothbeans, mungbean, muskmelon, orange, papaya, pigeonpeas, pomegranate, rice, and watermelon — contains exactly 100 observations, giving the dataset a perfectly balanced class structure that is well suited for both descriptive analysis and future classification modeling.

4. Methodology

4.1 Data Import

The dataset was imported into R using `read.csv()` from a local CSV file. Initial inspection with `head()`, `tail()`, `dim()`, `names()`, and `str()` confirmed a data frame of 2,200 rows and 8 columns, with seven numeric/integer fields and one character field for the crop label.

4.2 Data Exploration

The structure and class of each column were reviewed using `str()` and `sapply(..., class)`, confirming that `n`, `p`, and `k` were stored as integers, temperature, humidity, `ph`, and rainfall as numeric (double) values, and label as character text. A random sample of records was drawn using `sample_n()` to visually spot-check the data.

4.3 Data Cleaning

Column names were standardized to lowercase, syntactically valid names using `janitor::clean_names()`, converting the original headers (`N`, `P`, `K`, temperature, humidity, `ph`, rainfall, label) into consistent lowercase equivalents (`n`, `p`, `k`, temperature, humidity, `ph`, rainfall, label).

4.4 Missing Value Analysis

A column-wise missing-value audit was performed with `colSums(is.na(data))`. The result confirmed zero missing values across all eight columns, indicating a complete dataset that required no imputation.

4.5 Duplicate Check

Duplicate records were checked using `sum(duplicated(data))`, which returned zero, confirming that every row in the dataset is unique. The `distinct()` function was applied as a defensive step to guarantee row uniqueness prior to analysis.

4.6 Data Type Conversion

The label column was converted from character to factor type using `as.factor()`, enabling R to treat the 22 crop categories as a proper categorical variable with 22 levels for summary statistics, tabulation, and visualization.

4.7 Descriptive Statistics

Summary statistics were generated using `summary()`, `sapply(..., sd)`, `sapply(..., var)`, and a custom coefficient-of-variation calculation. Additional detailed distributional summaries were produced with `skimr::skim()` and `describe()`, providing percentiles, means, and compact inline histograms for every numeric variable.

4.8 Exploratory Data Analysis

Univariate distributions were visualized using `ggplot2` histograms (temperature, rainfall, soil pH) and a bar chart (crop distribution), while multivariate spread and outliers were examined using a combined boxplot of all seven numeric variables (via `pivot_longer()` reshaping). Bivariate relationships were explored using scatterplots with linear regression trend lines (`geom_smooth` with `method = 'lm'`) for temperature-rainfall, temperature-humidity, and pH-rainfall pairs.

4.9 Correlation Analysis

Pearson correlation coefficients were computed for all numeric variables using `cor()`, visualized as a color-coded correlation heatmap using `corrplot()`, and formally tested for statistical significance using `Hmisc::rcorr()`, which returns both correlation coefficients and their associated p-values for every variable pair.

5. Descriptive Statistics

The table below summarizes the central tendency, dispersion, and relative variability (coefficient of variation, CV) of each numeric variable in the dataset, based on the `summary()`, `sd()`, and CV calculations performed in R.

Variable	Mean	Median	Std. Dev.	Range	CV (%)
Nitrogen (n)	50.55	37.00	36.92	0 – 140	73.03
Phosphorus (p)	53.36	51.00	32.99	5 – 145	61.81
Potassium (k)	48.15	32.00	50.65	5 – 205	105.19
Temperature (°C)	25.62	25.60	5.06	8.83 – 43.68	19.77
Humidity (%)	71.48	80.47	22.26	14.26 – 99.98	31.15
Soil pH	6.47	6.43	0.77	3.50 – 9.94	11.96
Rainfall (mm)	103.46	94.87	54.96	20.21 – 298.56	53.12

5.1 Interpretation

Potassium (k) shows the highest relative variability of all variables (CV = 105.19%), meaning its standard deviation actually exceeds its mean — a signature of a right-skewed distribution with a long tail of high-potassium observations (visible later in the boxplot as a cluster of upper outliers). Nitrogen (CV = 73.03%) and phosphorus (CV = 61.81%) are similarly dispersed relative to their averages, reflecting the wide range of soil-nutrient requirements across 22 different crops rather than measurement noise.

By contrast, soil pH is by far the most stable variable (CV = 11.96%), clustering tightly around a mean of 6.47 — consistent with the fact that most cultivated crops in the dataset thrive within a comparatively narrow, near-neutral pH band. Temperature also shows relatively low variability (CV = 19.77%), with a mean of 25.62°C and a plausible tropical/subtropical range of roughly 9°C to 44°C. Humidity (CV = 31.15%) and rainfall (CV = 53.12%) fall in between, reflecting the diverse climatic zones represented by the 22 crop types, ranging from arid-tolerant crops such as mothbeans to high-rainfall crops such as rice, jute, and coconut.

The gap between mean and median for several variables (e.g., potassium: mean 48.15 vs. median 32.00; rainfall: mean 103.46 vs. median 94.87) confirms right-skewness — the mean is pulled upward by a minority of high-value observations, which is corroborated visually in the histograms and boxplots discussed in Section 6. Exact skewness and kurtosis coefficients were not part of the generated console output and are therefore not reported numerically here; distributional shape is instead assessed qualitatively from the histograms and boxplots in the following section.

6. Exploratory Data Analysis

This section presents and interprets each of the graphical outputs generated during exploratory analysis, covering the distribution of the target class and each numeric predictor, spread/outlier structure via boxplots, and bivariate relationships via scatterplots with fitted trend lines.

6.1 Distribution of Crop Types

Figure 1: Distribution of Crop Types

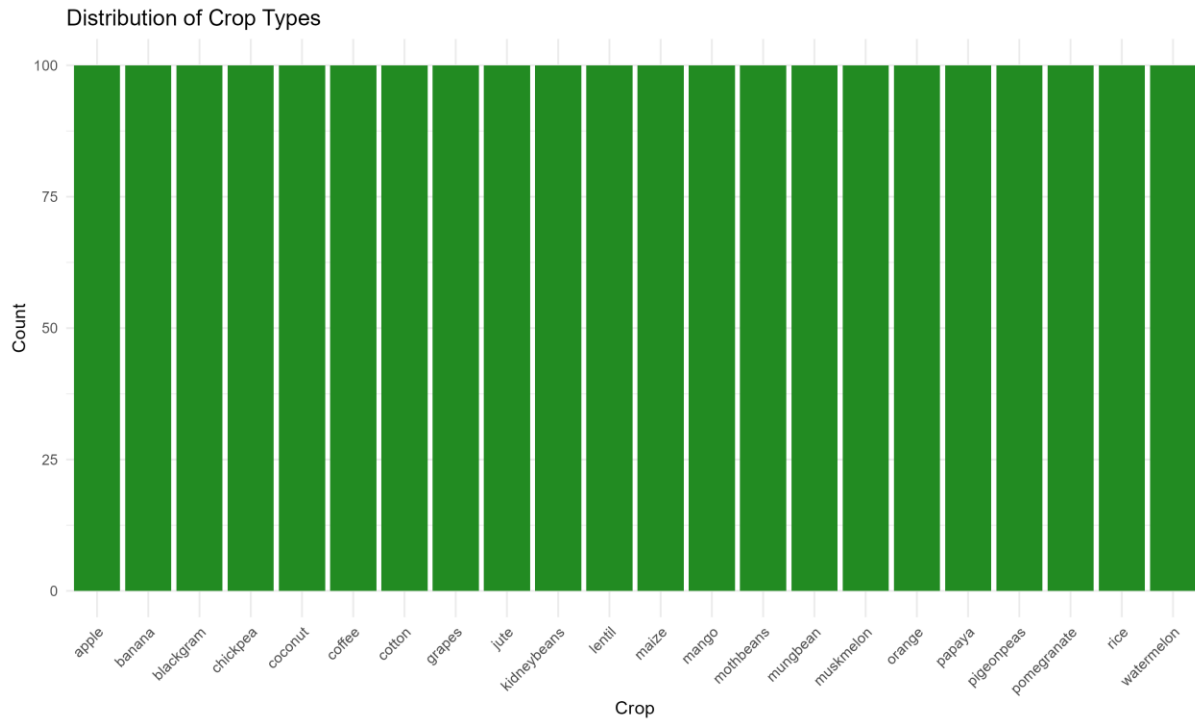


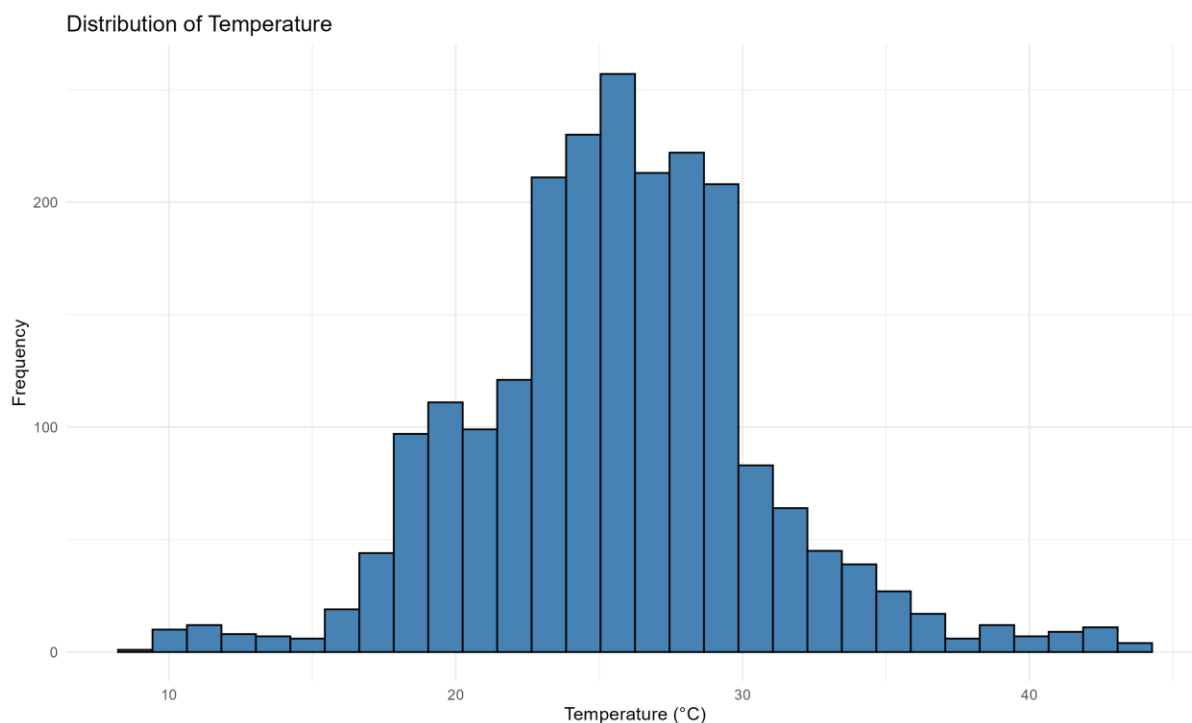
Figure 1. Distribution of Crop Types — generated in R using ggplot2.

The bar chart displays the frequency of each of the 22 crop labels in the dataset. Every bar reaches exactly 100, confirming that the dataset is perfectly balanced across all crop categories with no class dominating or under-represented.

Key Observations

- All 22 crops (apple, banana, blackgram, chickpea, coconut, coffee, cotton, grapes, jute, kidneybeans, lentil, maize, mango, mothbeans, mungbean, muskmelon, orange, papaya, pigeonpeas, pomegranate, rice, watermelon) appear with identical frequency ($n = 100$).
- The balanced design eliminates class-imbalance concerns for any future classification or crop-recommendation model built on this data.
- Because classes are equally represented, differences observed in numeric variables across the dataset are not artifacts of sampling frequency.

6.2 Histogram of Temperature

Figure 2: Histogram of Temperature*Figure 2. Histogram of Temperature — generated in R using ggplot2.*

The temperature histogram shows a broadly unimodal, moderately bell-shaped distribution centered in the low-to-mid twenties Celsius, with the tallest bars occurring between roughly 23°C and 29°C.

Key Observations

- The bulk of observations fall between approximately 18°C and 30°C, consistent with tropical and subtropical growing conditions.
- A secondary, lower-frequency cluster appears in the 8–17°C range, likely corresponding to cooler-climate crops such as chickpea or grapes.
- A thin right-hand tail extends toward the low 40s°C, indicating a small number of high-temperature observations rather than a strongly skewed bulk distribution.

6.3 Histogram of Rainfall

Figure 3: Histogram of Rainfall

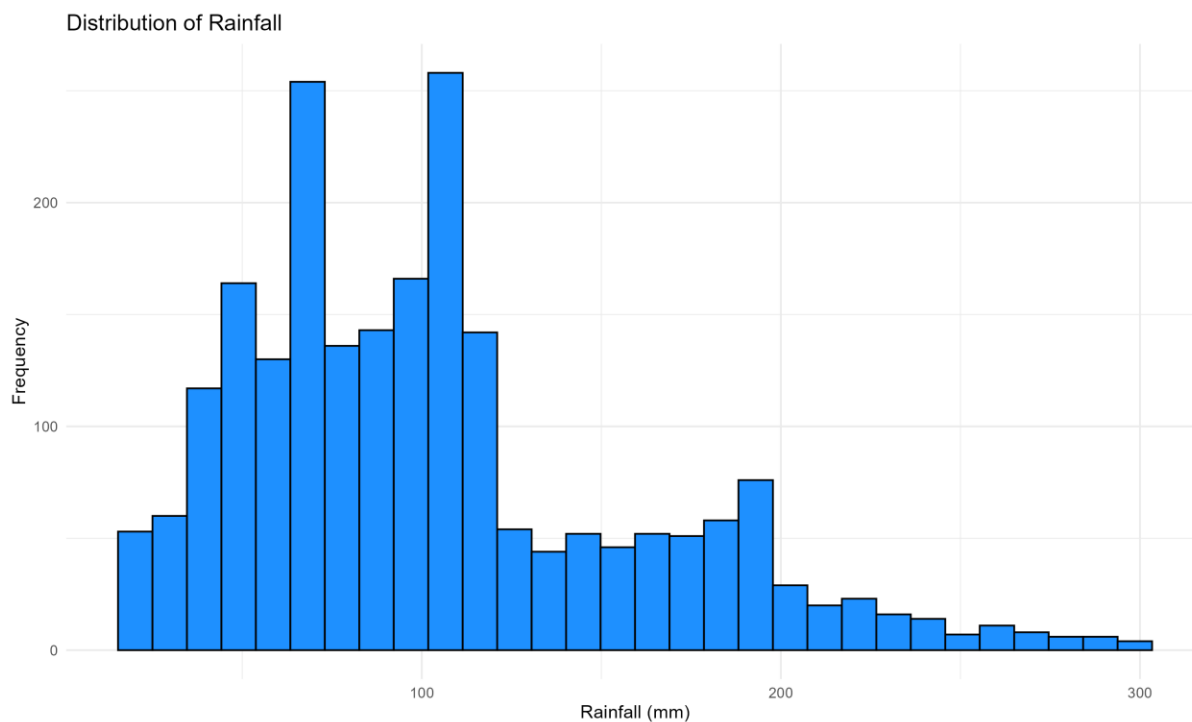


Figure 3. Histogram of Rainfall — generated in R using ggplot2.

Rainfall shows a distinctly right-skewed, multi-modal distribution. The tallest bars cluster between roughly 40 mm and 130 mm, but a long tail extends out to nearly 300 mm.

Key Observations

- The distribution is not smooth — several sharp peaks suggest distinct rainfall regimes tied to different crop-specific growing conditions rather than a single continuous climate gradient.
- A secondary elevated cluster appears around 170–200 mm, likely representing higher-rainfall crops such as rice, jute, papaya, or coconut.
- The long right tail toward 250–300 mm represents a minority of very high-rainfall observations, consistent with the elevated mean (103.46 mm) relative to the median (94.87 mm) noted in Section 5.

6.4 Histogram of Soil pH

Figure 4: Histogram of Soil pH

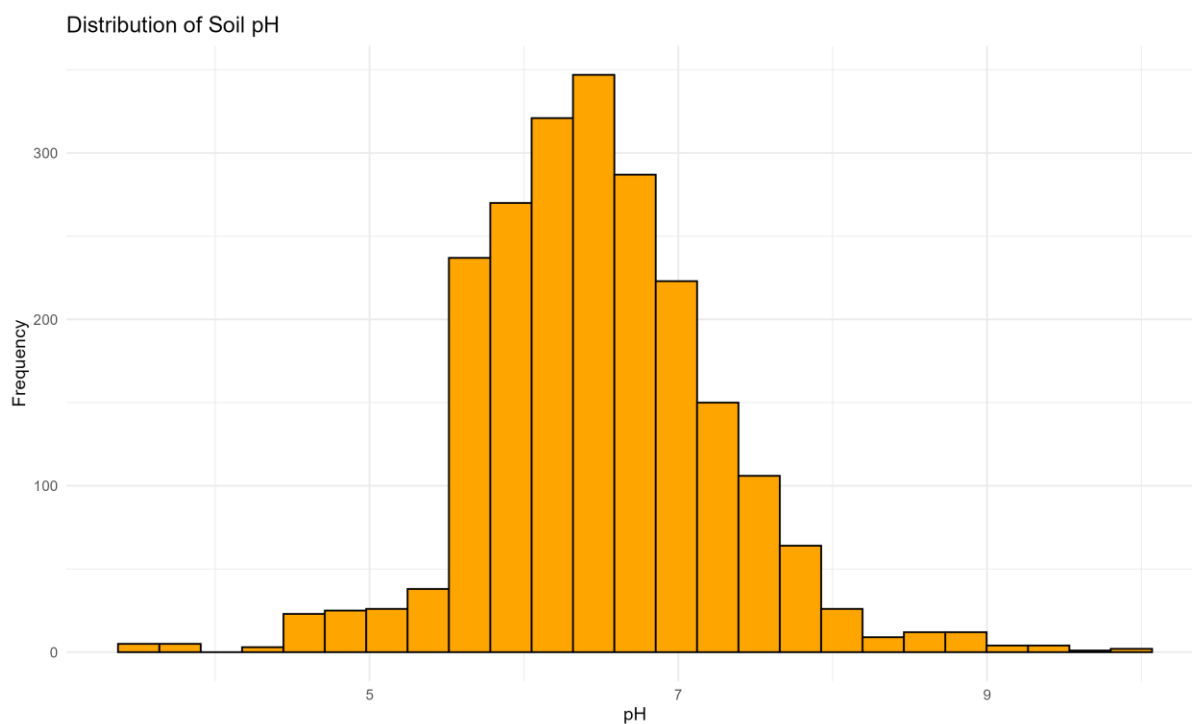


Figure 4. Histogram of Soil pH — generated in R using ggplot2.

Soil pH is the most symmetric and tightly concentrated of the numeric variables, forming a clear bell shape centered around pH 6–7 — the near-neutral range.

Key Observations

- The vast majority of observations fall between pH 5.5 and 7.5, aligning with the ideal pH range for most agricultural crops.
- A small number of low-frequency observations extend toward the acidic ($\text{pH} \approx 3.5\text{--}4.5$) and alkaline ($\text{pH} \approx 8.5\text{--}9.9$) extremes, representing crops with specialized soil tolerances.
- The tight, symmetric shape corresponds to the very low coefficient of variation (11.96%) computed in Section 5, confirming pH as the most stable variable in the dataset.

6.5 Boxplots of Numeric Variables

Figure 5: Boxplots of Numeric Variables

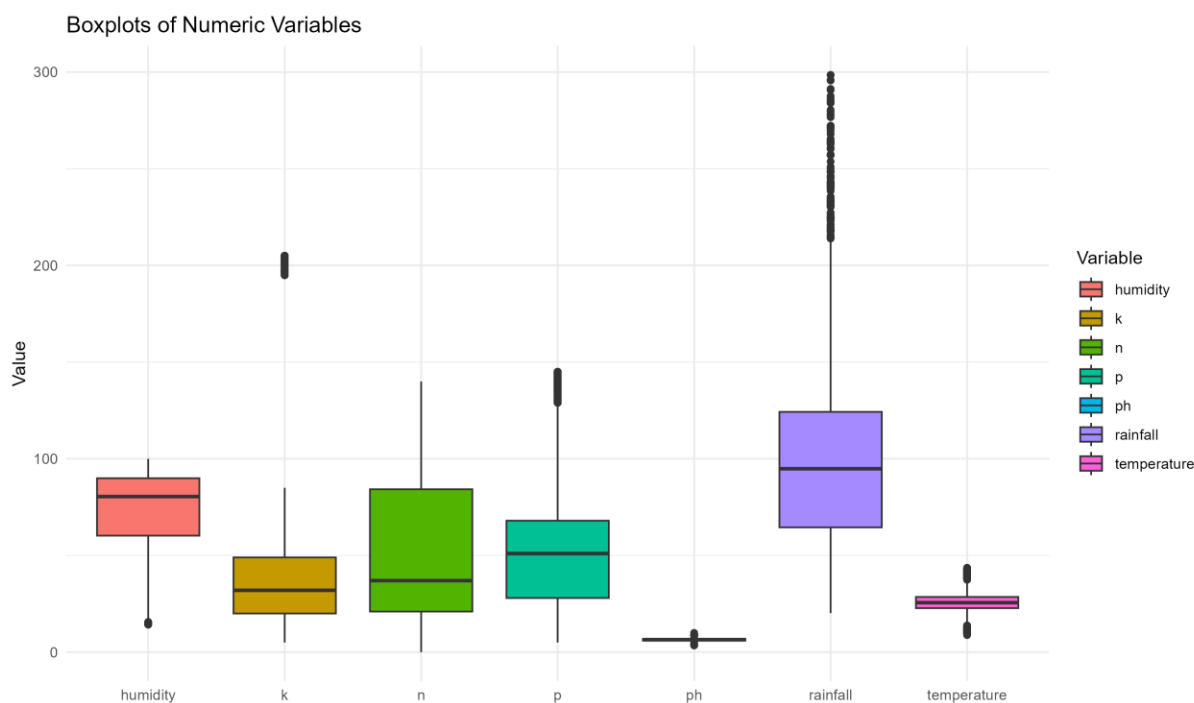


Figure 5. Boxplots of Numeric Variables — generated in R using ggplot2.

The combined boxplot (all seven numeric variables plotted on a shared scale after reshaping to long format) provides a compact view of spread, central tendency, and outlier structure across variables simultaneously.

Key Observations

- Rainfall shows the widest interquartile range and the most extensive set of upper outliers, with numerous points beyond 200 mm reaching close to 300 mm — visual confirmation of its right-skewed distribution.
- Potassium (k) displays a compact interquartile box near the lower end of its scale but a distinct, separated cluster of high outliers around 195–205, indicating a bimodal-like structure — likely a subset of crops (e.g., banana, grapes) requiring exceptionally high potassium levels.
- Phosphorus (p) shows a moderate number of upper outliers above roughly 130, echoing its strong correlation with potassium discussed in Section 7.
- Soil pH appears visually compressed on this shared axis because its natural range (3.5–9.9) is small relative to variables like rainfall and potassium — its low variability is consistent with the CV figures in Section 5.
- Humidity and temperature show comparatively narrow, symmetric boxes with few outliers, reflecting their moderate coefficients of variation.

6.6 Temperature vs Rainfall

Figure 6: Temperature vs Rainfall

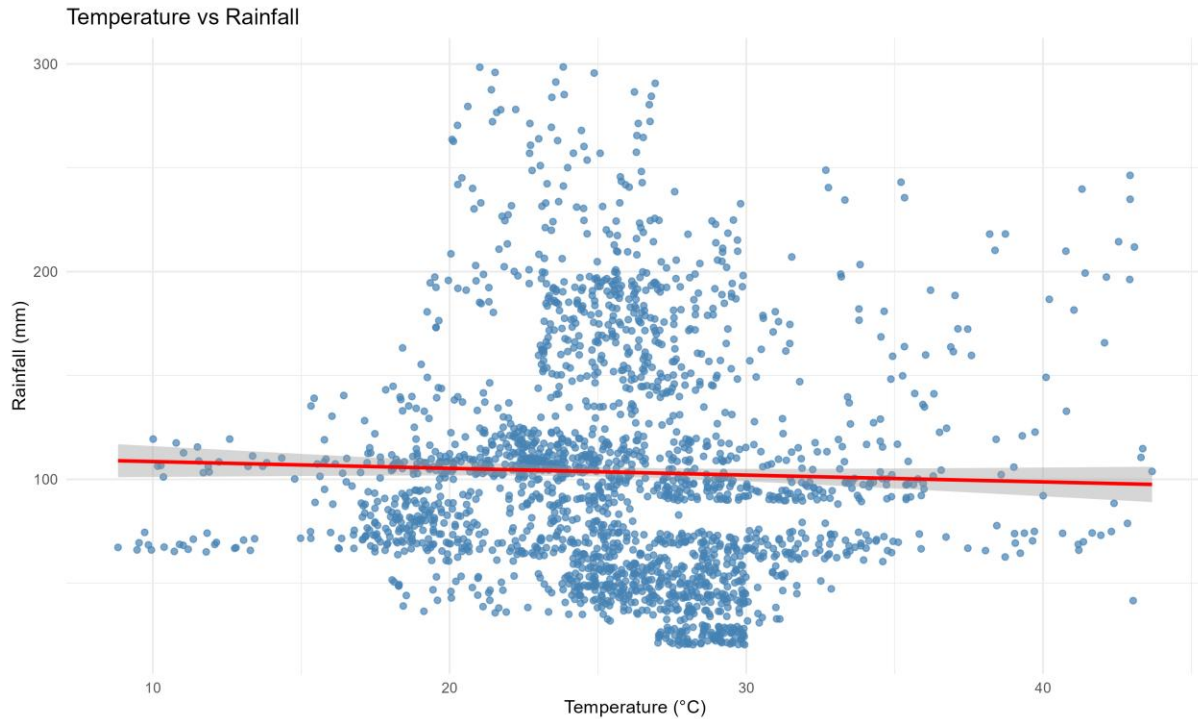


Figure 6. Temperature vs Rainfall — generated in R using ggplot2.

This scatterplot with a fitted linear regression line examines the relationship between temperature and rainfall. The fitted line is nearly flat with a very slight downward slope, and the surrounding confidence band is narrow, but the point cloud is extremely diffuse.

Key Observations

- The linear trend line indicates a negligible relationship, consistent with the near-zero Pearson correlation coefficient ($r = -0.03$) reported in Section 7.
- The point cloud shows several distinct vertical bands and clusters rather than a smooth continuous spread, again suggesting the influence of discrete crop-specific climate niches rather than a single continuous joint distribution.
- Practically, this indicates that temperature alone is a poor linear predictor of rainfall in this dataset — the two variables vary largely independently across crop types.

6.7 Temperature vs Humidity

Figure 7: Temperature vs Humidity

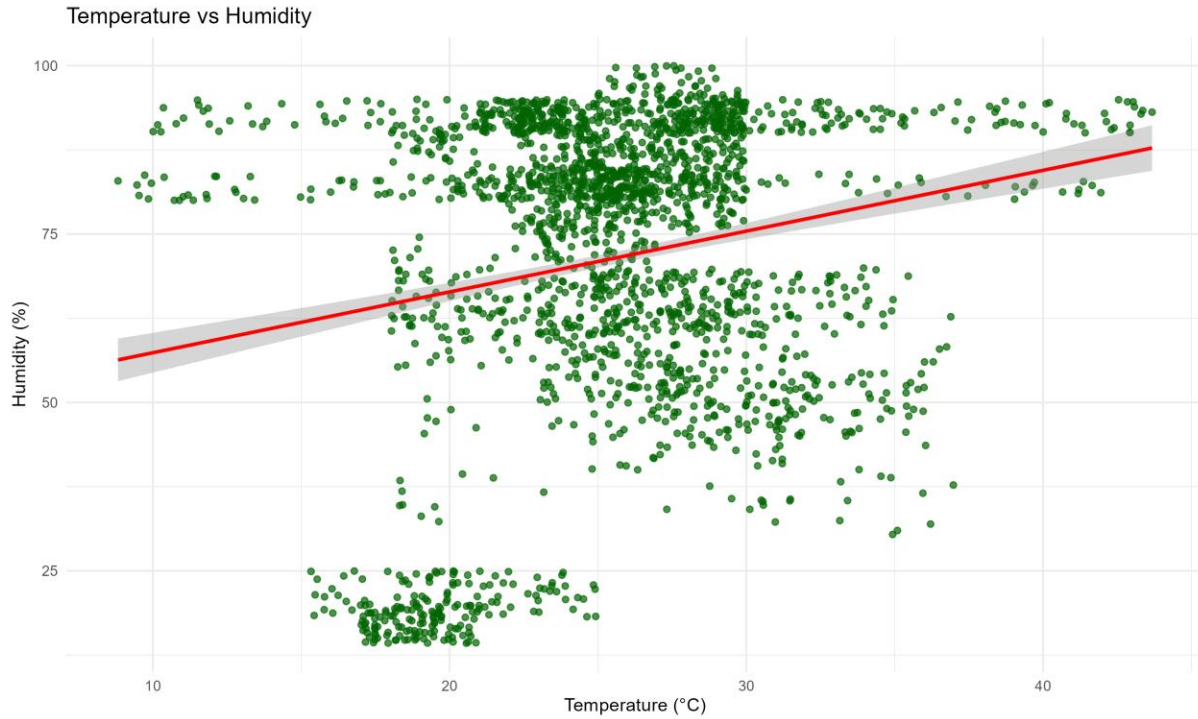


Figure 7. Temperature vs Humidity — generated in R using ggplot2.

This scatterplot shows a positive linear trend line, with humidity generally increasing as temperature increases, though the underlying data show substantial scatter and several distinct horizontal bands.

Key Observations

- The fitted regression line has a visibly positive slope, matching the weak-to-moderate positive Pearson correlation ($r = 0.21$) identified in Section 7 — the strongest of the three scatterplot relationships examined.
- Several horizontal 'stripe' clusters are visible (e.g., near 80–95% humidity, 55–70% humidity, and a low band near 15–25% humidity), again reflecting distinct crop-specific climate profiles rather than one continuous relationship.
- The relationship, while statistically detectable, remains too weak to be practically predictive on its own — humidity is jointly influenced by many additional agro-climatic factors not captured in a single bivariate view.

6.8 Soil pH vs Rainfall

Figure 8: Soil pH vs Rainfall

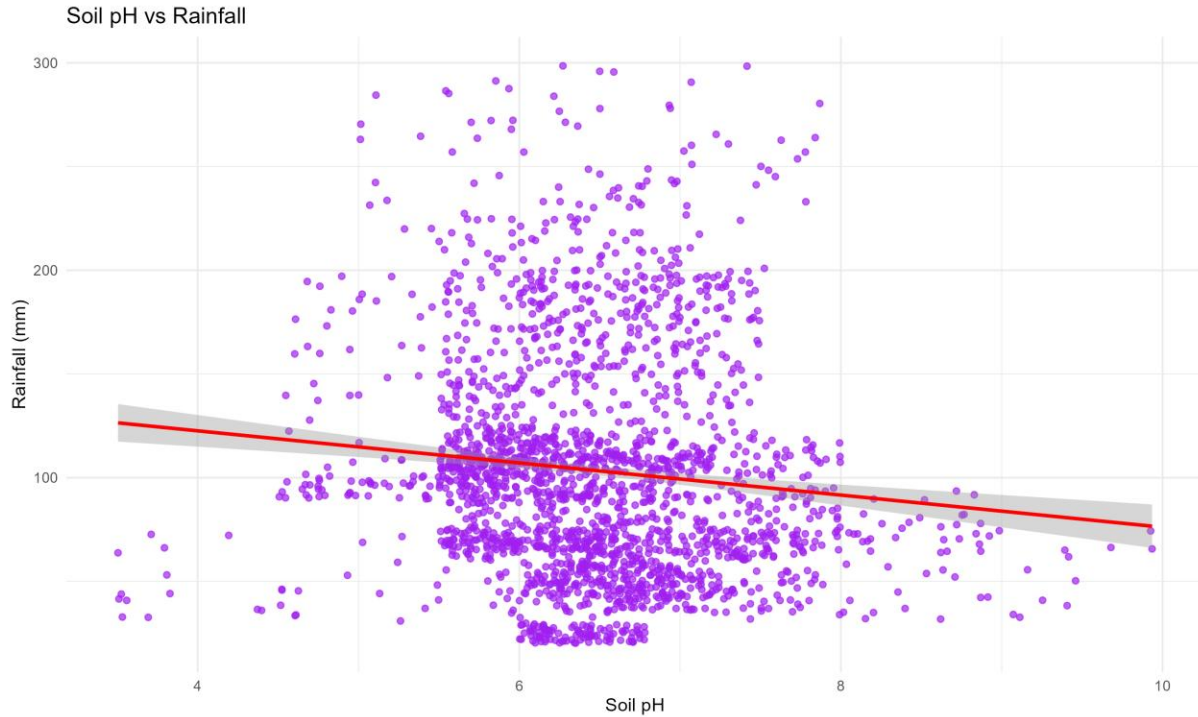


Figure 8. Soil pH vs Rainfall — generated in R using ggplot2.

This scatterplot examines the relationship between soil pH and rainfall, showing a fitted regression line with a mild negative slope, indicating that higher-rainfall observations tend to be associated with slightly more acidic soils.

Key Observations

- The negative slope is consistent with the weak negative Pearson correlation ($r = -0.11$) reported in Section 7, which is statistically significant but practically modest in magnitude.
- The point cloud is dense and diffuse across the full pH range, with a particularly heavy concentration between pH 5.5–7.5, mirroring the histogram distribution described in Section 6.4.
- The wide scatter around the trend line indicates that pH explains only a small fraction of the variance in rainfall, reinforcing that these two variables act largely as independent predictors of crop suitability.

7. Correlation Analysis

7.1 Correlation Matrix

Pearson correlation coefficients were computed for all seven numeric variables. The full matrix (rounded to two decimal places) is reproduced below.

	n	p	k	temp	humidity	ph	rainfall
n	1.00	-0.23	-0.14	0.03	0.19	0.10	0.06
p	-0.23	1.00	0.74	-0.13	-0.12	-0.14	-0.06
k	-0.14	0.74	1.00	-0.16	0.19	-0.17	-0.05
temperature	0.03	-0.13	-0.16	1.00	0.21	-0.02	-0.03
humidity	0.19	-0.12	0.19	0.21	1.00	-0.01	0.09
ph	0.10	-0.14	-0.17	-0.02	-0.01	1.00	-0.11
rainfall	0.06	-0.06	-0.05	-0.03	0.09	-0.11	1.00

7.2 Correlation Heatmap

Figure 9: Correlation Heatmap of Numeric Variables

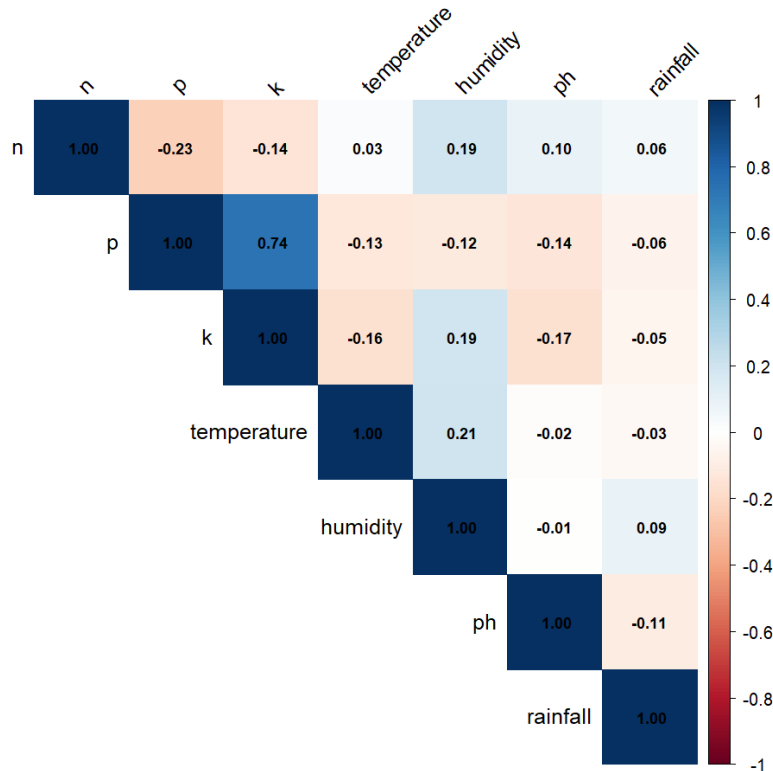


Figure 9. Pearson correlation heatmap generated using `corrplot()` (upper triangle, color + coefficient labels).

The heatmap visually confirms the numeric matrix: the phosphorus–potassium cell is the single strongly saturated blue cell off the diagonal ($r = 0.74$), while every other off-diagonal cell is a pale shade close to white, reflecting correlations under $|0.25|$ in magnitude.

7.3 Correlation Significance Testing

To confirm which correlations are statistically meaningful rather than due to chance, Pearson correlation coefficients were re-computed with associated p-values using `Hmisc::rcorr()`. A selection of the most relevant pairs is shown below ($p < 0.05$ considered statistically significant).

Variable Pair	r	p-value	Significance
n – p	-0.231	< 0.001	Significant
n – k	-0.141	< 0.001	Significant
n – temperature	0.027	0.214	Not significant
n – humidity	0.191	< 0.001	Significant
p – k	0.736	< 0.001	Significant (strongest)
temperature – humidity	0.205	< 0.001	Significant
temperature – ph	-0.018	0.404	Not significant
humidity – ph	-0.008	0.691	Not significant
ph – rainfall	-0.109	< 0.001	Significant
temperature – rainfall	-0.030	0.159	Not significant

7.4 Discussion

Strong Positive Correlations

The only strong correlation in the dataset is between phosphorus (p) and potassium (k), at $r = 0.74$ ($p < 0.001$). This was also the sole pair flagged by the analysis when filtering for $|r| > 0.70$. Agronomically, this is a highly plausible and expected relationship: phosphorus and potassium are frequently co-applied through compound fertilizers (e.g., NPK blends) and often occur together in similar soil types, so soils rich in one tend to be rich in the other.

Weak Correlations

Every other variable pair in the dataset exhibits a weak correlation, with magnitudes at or below 0.23. Notably, nitrogen is weakly negatively correlated with both phosphorus ($r = -0.23$) and

potassium ($r = -0.14$) — a mild trade-off pattern rather than a strong inverse relationship. Temperature shows its strongest (though still weak) association with humidity ($r = 0.21$), and soil pH is weakly negatively associated with both potassium ($r = -0.17$) and rainfall ($r = -0.11$).

Statistically Non-Significant Relationships

Several pairs — nitrogen vs. temperature ($p = 0.214$), temperature vs. soil pH ($p = 0.404$), humidity vs. soil pH ($p = 0.691$), and temperature vs. rainfall ($p = 0.159$) — show no statistically significant linear relationship at the conventional 0.05 threshold. This indicates that, within this dataset, temperature and pH behave essentially independently of nitrogen levels and of each other.

Practical Agricultural Implications

- Because most environmental variables are weakly and often independently related, each one is likely to carry unique, non-redundant information for a future crop-recommendation or classification model — multicollinearity is not a major concern except for the p–k pair.
- The strong p–k relationship suggests that soil-testing programs and fertilizer-recommendation systems could reasonably treat phosphorus and potassium as a linked pair when advising on soil amendments.
- The lack of a strong temperature–rainfall relationship in this dataset means crop climate suitability should be assessed using temperature and rainfall as separate, complementary criteria rather than inferring one from the other.
- The near-independence of soil pH from temperature and humidity suggests pH management (e.g., liming or sulfur amendment) can be approached as a distinct soil-chemistry lever, unconstrained by the prevailing climate.

8. Key Findings

- The dataset contains 2,200 records across 22 crop types, each represented by exactly 100 observations — a perfectly balanced class structure.
- No missing values and no duplicate rows were found in any of the 8 columns, indicating high underlying data quality.
- Soil pH is the most stable numeric variable (CV = 11.96%), clustering tightly around a mean of 6.47 in a near-bell-shaped, symmetric distribution.
- Potassium (k) is the most variable numeric feature (CV = 105.19%), with its standard deviation exceeding its mean — a hallmark of strong right-skew driven by a subset of very high-potassium crops.
- Rainfall displays a distinctly right-skewed, multi-modal distribution, with a long tail of observations extending to almost 300 mm.
- Temperature is roughly bell-shaped and concentrated between 18°C and 30°C, consistent with tropical/subtropical growing conditions, with a smaller secondary cluster in cooler ranges.
- The boxplot of all numeric variables reveals substantial outlier activity in rainfall, potassium, and phosphorus, but comparatively little in temperature, humidity, and pH.
- Phosphorus and potassium show the only strong correlation in the dataset ($r = 0.74$, $p < 0.001$), reflecting their common co-occurrence in fertilizer application and soil composition.
- All other variable pairs are weakly correlated ($|r| \leq 0.23$), suggesting most environmental features act largely independently of one another.
- Temperature and humidity show the strongest positive relationship among the climate variables ($r = 0.21$, $p < 0.001$), though the effect remains modest in absolute terms.
- Soil pH is weakly negatively correlated with rainfall ($r = -0.11$, $p < 0.001$) and potassium ($r = -0.17$, $p < 0.001$).
- Several relationships — nitrogen vs. temperature, temperature vs. pH, humidity vs. pH, and temperature vs. rainfall — are not statistically significant at the 0.05 level, indicating no detectable linear association.
- Nitrogen shows a mild negative association with both phosphorus and potassium ($r = -0.23$ and $r = -0.14$ respectively), suggesting a modest nutrient trade-off pattern across crop types rather than co-occurrence.
- The overall low degree of multicollinearity among predictors (aside from p–k) suggests the dataset is well suited to multivariate statistical modeling and machine-learning classification without extensive feature-reduction requirements.

- The combination of a clean, balanced dataset with clearly interpretable variable relationships makes this dataset a strong candidate for developing a data-driven crop recommendation system.

9. Conclusion

This statistical analysis of the Crop Recommendation dataset demonstrates that a rigorous, well-structured exploratory workflow can extract substantial agronomic insight from a relatively compact set of environmental variables. The dataset's data quality is excellent — free of missing values and duplicate records — which allowed the analysis to move directly from import into descriptive and inferential statistics without requiring a cleaning or imputation phase.

The statistical profile of the seven numeric variables reveals a mix of stable, tightly distributed features (most notably soil pH) and highly variable, right-skewed features (most notably potassium and rainfall), each shaped by the diversity of the 22 crop types represented. Exploratory visualizations — histograms, boxplots, and a bar chart of crop frequency — confirmed both the balanced class design of the target variable and the distinct climatic and nutrient niches occupied by different crops.

Correlation analysis showed that, aside from the strong and agronomically sensible phosphorus–potassium relationship, most variable pairs are weakly correlated or statistically independent. This is a favorable property for downstream analysis: it indicates that each environmental variable contributes largely unique, non-redundant information toward distinguishing crop suitability, which is precisely the kind of structure that supports robust classification and recommendation modeling.

More broadly, this project illustrates the value of exploratory data analysis as a precursor to any predictive modeling effort. By characterizing distributions, identifying outliers, and quantifying inter-variable relationships before modeling, analysts can make more informed choices about feature engineering, model selection, and the interpretation of downstream results. In the agricultural context specifically, these findings can support soil-testing programs, fertilizer-recommendation systems, and precision-agriculture tools aimed at matching crops to the environmental conditions in which they will perform best.

10. Future Scope

Building on this exploratory foundation, several extensions could deepen the analytical and practical value of this work:

- Predictive Modeling: Develop supervised classification models (e.g., logistic regression, random forest, gradient boosting, or neural networks) to predict the most suitable crop label from the seven environmental features.

- **Machine Learning Pipelines:** Implement automated feature scaling, train/test splitting, and cross-validation to benchmark multiple algorithms on classification accuracy, precision, recall, and F1-score.
- **Crop Recommendation Systems:** Package the trained model into an interactive tool (e.g., a Shiny app or web API) that accepts soil and climate readings and returns a ranked list of recommended crops.
- **Yield Prediction:** Extend the dataset with historical yield figures to move beyond crop classification toward quantitative yield forecasting under varying environmental conditions.
- **Precision Agriculture:** Integrate geospatial and sensor-derived data (satellite imagery, IoT soil sensors, weather station feeds) to enable location-specific, real-time crop and fertilizer recommendations.
- **Feature Engineering:** Explore interaction terms and non-linear transformations (e.g., N:P:K ratios) that may better capture agronomic relationships than the raw linear correlations examined here.
- **Outlier and Subgroup Analysis:** Conduct crop-level (grouped) descriptive statistics to characterize the specific environmental niche of each of the 22 crops individually, rather than analyzing the dataset only in aggregate.

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